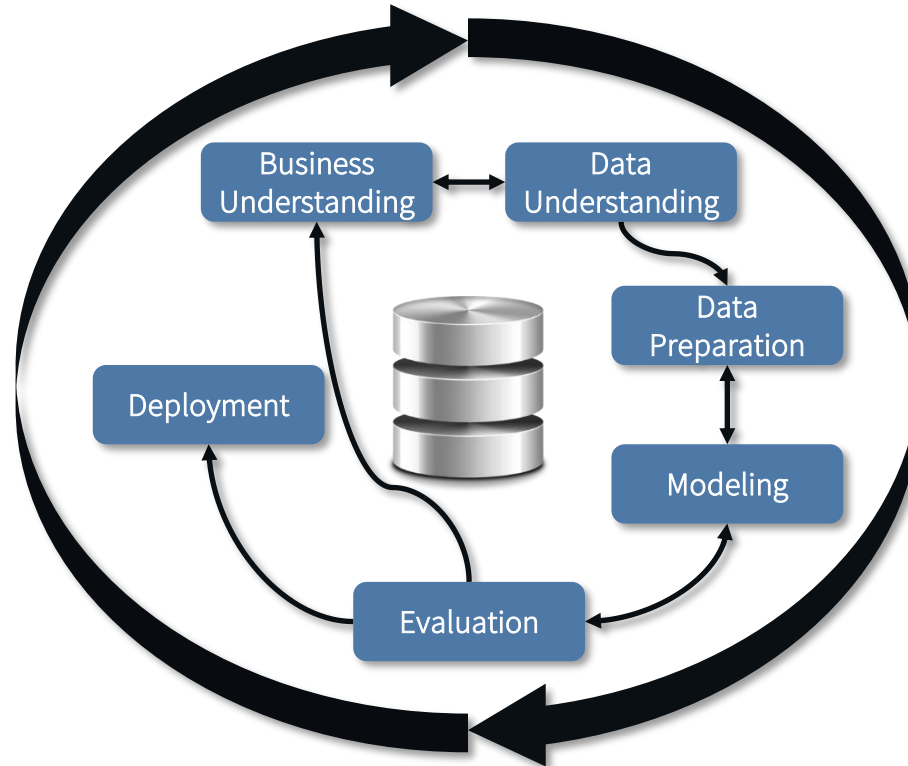


Project Setup for Reproduceability – Checklist

1. Clear analysis process (e.g. CRISP DM)
2. Self-contained project folder, well structured
3. Raw data is immutable!
4. Raw data + code = replicable results
5. Project-specific virtual environment (managed with uv)
6. README.md as entry point for good documentation
7. Specifying seed / random_state / ... for random processes
8. Sufficient trials for non-deterministic processes
9. Software-Engineering best practices (see next slide)

CRISP DM* – Data Science Lifecycle

* Cross Industry Standard Process for Data Mining



Software Engineering Best Practices

1. Version control with git / GitHub
2. Project-Package for reusable classes and functions
3. Log files (logging package)
4. Tests (pytest)
5. Linting (ruff)
6. Pre-commit hooks for automated checks before commit to repo
7. Optional: GitHub Actions / Continuous Integration Pipelines for automated checks after push to repo
8. Optional: Documentation with sphinx